

Hence

$$\int C'(x) dx = \int (30 - 0.02x) dx$$

and

$$C(x) = 30x - 0.01x^2 + K$$

for some K . Letting $x = 1$ and using $C(1) = 35$, we obtain

$$35 = 30 - 0.01 + K, \text{ or } K = 5.01.$$

Consequently,

$$C(x) = 30x - 0.01x^2 + 5.01.$$

In particular, the cost of producing 100 units is

$$C(100) = 3000 - 100 + 5.01 = \$2905.01.$$

EXERCISES 5.1

Exer. 1–42: Evaluate.

$$1 \int (4x + 3) dx$$

$$2 \int (4x^2 - 8x + 1) dx$$

$$3 \int (9t^2 - 4t + 3) dt$$

$$4 \int (2t^3 - t^2 + 3t - 7) dt$$

$$5 \int \left(\frac{1}{z^3} - \frac{3}{z^2} \right) dz$$

$$6 \int \left(\frac{4}{z^7} - \frac{7}{z^4} + z \right) dz$$

$$7 \int \left(3\sqrt{u} + \frac{1}{\sqrt{u}} \right) du$$

$$8 \int (\sqrt{u^3} - \frac{1}{2}u^{-2} + 5) du$$

$$9 \int (2v^{5/4} + 6v^{1/4} + 3v^{-4}) dv$$

$$10 \int (3v^5 - v^{5/3}) dv$$

$$11 \int (3x - 1)^2 dx$$

$$12 \int \left(x - \frac{1}{x} \right)^2 dx$$

$$13 \int x(2x + 3) dx$$

$$14 \int (2x - 5)(3x + 1) dx$$

$$15 \int \frac{8x - 5}{\sqrt[3]{x}} dx$$

$$16 \int \frac{2x^2 - x + 3}{\sqrt{x}} dx$$

$$17 \int \frac{x^3 - 1}{x - 1} dx, \quad x \neq 1$$

$$18 \int \frac{x^3 + 3x^2 - 9x - 2}{x - 2} dx, \quad x \neq 2$$

$$19 \int \frac{(t^2 + 3)^2}{t^6} dt$$

$$20 \int \frac{(\sqrt{t} + 2)^2}{t^3} dt$$

$$21 \int \frac{3}{4} \cos u du$$

$$22 \int -\frac{1}{5} \sin u du$$

$$23 \int \frac{7}{\csc x} dx$$

$$24 \int \frac{1}{4 \sec x} dx$$

$$25 \int (\sqrt{t} + \cos t) dt$$

$$26 \int (\sqrt[3]{t^2} - \sin t) dt$$

$$27 \int \frac{\sec t}{\cos t} dt$$

$$29 \int (\csc v \cot v \sec v) dv$$

$$31 \int \frac{\sec w \sin w}{\cos w} dw$$

$$33 \int \frac{(1 + \cot^2 z) \cot z}{\csc z} dz$$

$$35 \int D_x \sqrt{x^2 + 4} dx$$

$$37 \int \frac{d}{dx} (\sin \sqrt[3]{x}) dx$$

$$39 \int D_x \int (x^3 \sqrt{x - 4}) dx$$

$$41 \int \frac{d}{dx} \cot x^3 dx$$

$$28 \int \frac{1}{\sin^2 t} dt$$

$$30 \int (4 + 4 \tan^2 v) dv$$

$$32 \int \frac{\csc w \cos w}{\sin w} dw$$

$$34 \int \frac{\tan z}{\cos z} dz$$

$$36 \int D_x \sqrt[3]{x^3 - 8} dx$$

$$38 \int \frac{d}{dx} (\sqrt{\tan x}) dx$$

$$40 \int D_x \int (x^4 \sqrt[3]{x^2 + 9}) dx$$

$$42 \int \frac{d}{dx} \cos \sqrt{x^2 + 1} dx$$

Exer. 43–48: Evaluate the integral if a and b are constants.

$$43 \int a^2 dx$$

$$44 \int ab dx$$

$$45 \int (at + b) dt$$

$$46 \int \left(\frac{a}{b^2} t \right) dt$$

$$47 \int (a + b) du$$

$$48 \int (b - a^2) du$$

Exer. 49–56: Solve the differential equation subject to the given conditions.

$$49 \quad f'(x) = 12x^2 - 6x + 1; \quad f(1) = 5$$

$$50 \quad f'(x) = 9x^2 + x - 8; \quad f(-1) = 1$$

$$51 \quad \frac{dy}{dx} = 4x^{1/2};$$

$$y = 21 \text{ if } x = 4$$

- 52 $\frac{dy}{dx} = 5x^{-1/3}$; $y = 70$ if $x = 27$
- 53 $f''(x) = 4x - 1$; $f'(2) = -2$; $f(1) = 3$
- 54 $f''(x) = 6x - 4$; $f'(2) = 5$; $f(2) = 4$
- 55 $\frac{d^2y}{dx^2} = 3 \sin x - 4 \cos x$; $y = 7$ and $y' = 2$ if $x = 0$
- 56 $\frac{d^2y}{dx^2} = 2 \cos x - 5 \sin x$; $y = 2 + 6\pi$ and $y' = 3$ if $x = \pi$

Exer. 57–58: If a point is moving on a coordinate line with the given acceleration $a(t)$ and initial conditions, find $s(t)$.

- 57 $a(t) = 2 - 6t$; $v(0) = -5$; $s(0) = 4$
- 58 $a(t) = 3t^2$; $v(0) = 20$; $s(0) = 5$
- 59 A projectile is fired vertically upward from ground level with a velocity of 1600 ft/sec. Disregarding air resistance, find
- (a) its distance $s(t)$ above ground at time t
 - (b) its maximum height
- 60 An object is dropped from a height of 1000 feet. Disregarding air resistance, find
- (a) the distance it falls in t seconds
 - (b) its velocity at the end of 3 seconds
 - (c) when it strikes the ground
- 61 A stone is thrown directly downward from a height of 96 feet with an initial velocity of 16 ft/sec. Find
- (a) its distance above the ground after t seconds
 - (b) when it strikes the ground
 - (c) the velocity at which it strikes the ground
- 62 A gravitational constant for objects near the surface of the moon is 5.3 ft/sec².
- (a) If an astronaut on the moon throws a stone directly upward with an initial velocity of 60 ft/sec, find its maximum altitude.
 - (b) If, after returning to Earth, the astronaut throws the same stone directly upward with the same initial velocity, find the maximum altitude.
- 63 If a projectile is fired vertically upward from a height of s_0 feet above the ground with a velocity of v_0 ft/sec, prove that if air resistance is disregarded, its distance $s(t)$ above the ground after t seconds is given by $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$, where g is a gravitational constant.
- 64 A ball rolls down an inclined plane with an acceleration of 2 ft/sec².
- (a) If the ball is given no initial velocity, how far will it roll in t seconds?
 - (b) What initial velocity must be given for the ball to roll 100 feet in 5 seconds?
- 65 If an automobile starts from rest, what constant acceleration will enable it to travel 500 feet in 10 seconds?
- 66 If a car is traveling at a speed of 60 mi/hr, what constant (negative) acceleration will enable it to stop in 9 seconds?
- 67 A small country has natural gas reserves of 100 billion ft³. If $A(t)$ denotes the total amount of natural gas consumed after t years, then dA/dt is the rate of consumption. If the rate of consumption is predicted to be $5 + 0.01t$ billion ft³/year, in approximately how many years will the country's natural gas reserves be depleted?
- 68 Refer to Exercise 67. Based on U.S. Department of Energy statistics, the rate of consumption of gasoline in the United States (in billions of gallons per year) is approximated by $dA/dt = 2.74 - 0.11t - 0.01t^2$, with $t = 0$ corresponding to the year 1980. Estimate the number of gallons of gasoline consumed in the United States between 1980 and 1984.
- 69 A sportswear manufacturer determines that the marginal cost in dollars of producing x warm-up suits is given by $20 - 0.015x$. If the cost of producing one suit is \$25, find the cost function and the cost of producing 50 suits.
- 70 If the marginal cost function of a product is given by $2/x^{1/3}$ and if the cost of producing 8 units is \$20, find the cost function and the cost of producing 64 units.

5.2 CHANGE OF VARIABLES IN INDEFINITE INTEGRALS

The formulas for indefinite integrals in Table (5.4) are limited in scope because we cannot use them directly to evaluate integrals such as

$$\int \sqrt{5x+7} \, dx \quad \text{or} \quad \int \cos 4x \, dx.$$

In this section we shall develop a simple but powerful method for changing the variable of integration so that these integrals (and many others) can be evaluated by using the formulas in Table (5.4).